ELENI KAMBOURIS

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SUMMARY

Aerospace Engineering M.S. graduate with experience in aircraft systems, performance analysis, and multidisciplinary engineering across design, integration, and testing. Background includes hands-on aircraft development, working across aerodynamics, propulsion, and structural systems to support overall vehicle performance. Proficient in MATLAB and engineering analysis. Experience in modeling, simulation, and data-driven evaluation. Familiar with aircraft systems, sensors, and control-related concepts. Experience supporting system integration and experimental testing. Strong collaborative skills and ability to work across disciplines in fast-paced environments. Motivated to contribute to the development, integration, and testing of flight control and aircraft systems. US Citizen.

Security Clearances: TS/SCI with CI Polygraph (Fully adjudicated 20 December 2023).

EXPERIENCE

Department of Defense, USTRANSCOM: 4-Star Combatant Command

Analysis Career Field Intern

Rest of World Branch, Eurasia (Full-Time) — May 2025 – August 2025

- Authored two transportation analysis products on topics of key importance in Eurasia.
- Briefed analytic findings to the Deputy J3, a 1-star General.
- Co-authored a third, substantial product for executive attention published in September 2025.

Vendor Threat Intelligence Branch (Full-Time) — May 2024 – August 2024

- Produced an analytic highlight for publication that facilitated new communication between divisions.
- Supported vendor threat intelligence operations by analyzing assigned topics and contributing to internal analytic products.
- Assisted in training and quality control efforts for mission managers, improving analytic consistency.

Washington University, Design Build Fly (DBF), St. Louis, MO

Technical President — June 2024 – June 2025

- Led aircraft configuration, aerodynamic, and structural design efforts, achieving 14th-place technical acceptance and 26th-place design report ranking.
- Directed 6 cross-functional teams through aircraft conceptual design, configuration, fabrication, and flight testing for a national RC aircraft competition, achieving a 35th-place overall finish out of 115 teams.

Manufacturing Team Co-Lead — September 2022 – June 2024

- Supervised and trained 8 members in structural and composite aircraft manufacturing to produce competition-ready aircraft.
- Trained members on component construction from SolidWorks model to DXF files to laser cut components, jigs, and molds.
- Improved manufacturing processes through materials testing and fabrication method evaluation, contributing to a 3rd-place report and 11th-place overall finish (2022); Top 20 report, top-third finish (2023).

Washington University, Flores Research Group, St. Louis, MO

Undergraduate Researcher – Bulk Metallic Glasses — September 2023 – June 2024

- Utilized direct laser deposition additive manufacturing to construct walls of glass-forming powders on substrates at varying processing parameters such as laser power, travel speed, and powder feed rate.
- Characterized the structure of additively manufactured walls using XRD, DSC, microscopy and other techniques to evaluate presence of amorphous structure.

Teaching Assistant Experience: Fluids and Heat Transfer Lab, Senior Design, Introduction to Mechanical Engineering.

EDUCATION

Washington University in St. Louis, McKelvey School of Engineering

- Master of Science in Aerospace Engineering — Conferred December 2025
- Bachelor of Science in Mechanical Engineering, Minor in Materials Science — Conferred May 2025
- *Relevant Coursework:* CAD, Aircraft Performance, Design of Thermal Systems, Rotary-Wing Systems, CFD, Aerodynamics, Thermodynamics, Propulsion, Technical Writing

TECHNICAL SKILLS

- *Software*: MATLAB, SolidWorks, Python, Microsoft Office, Google Suites
- *Other*: CAD, Engineering Sketching/Drawing, GD & T, Lathe, Drill Press, Grinder, Mill, Sand Blaster, Band Saw, Belt Sander, Laser Cutting, Hot Wire Foam Cutter, Instron, MonoKote, Oscilloscope, DMM, Function Generator, Carbon Fiber and Fiberglass Layups, Partial-Face Respirator Use, Mold/Plug Making

AWARDS & OTHER

Pi Tau Sigma (Mechanical Engineering Honors Society); Linda Kral Prize for Outstanding Achievement in Aerospace Engineering (2025); WashU Dean's List (7 Semesters); Maryland State Governor's Merit (2021); MCPS Jazz Soloist Award (2019); AIAA Member. **In Progress:** CSWA (Expected Completion June 2026).